

Intro to Neural Networks & Mechanistic Interpretability

Maths, Physics, and AI Alignment

Julian Schulz

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1. **Machine Learning**
 - 1.1 Concepts
 - 1.2 Architectures
2. **Mechanistic Interpretability**
3. **Features**
4. **Circuits**
5. **Open Problems**

Machine Learning

1. **Machine Learning**

1.1 **Concepts** ←

1.2 Architectures

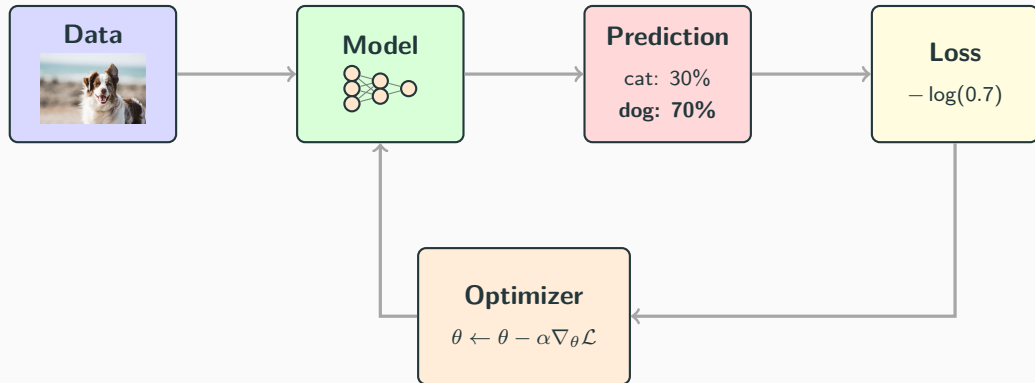
2. Mechanistic Interpretability

3. Features

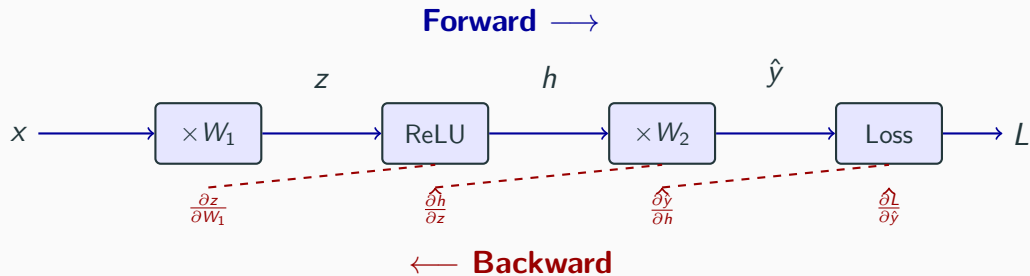
4. Circuits

5. Open Problems

The Training Loop



Forward Pass / Backward Pass



Chain rule:

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h} \cdot \frac{\partial h}{\partial z} \cdot \frac{\partial z}{\partial W_1}$$

Train Loss vs. Test Loss

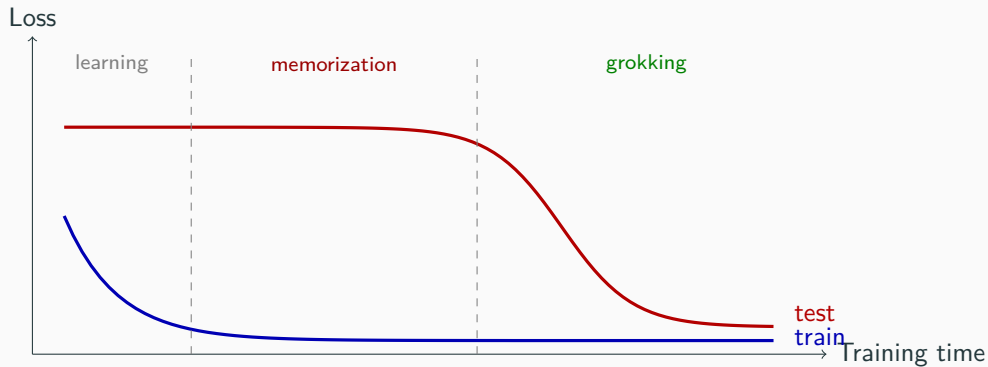


Double Descent



“Deep Double Descent: Where Bigger Models and More Data Can Hurt” [Nakkiran et al., 2019]

Grokking



"Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets" [Power et al., 2022]

Parameter / Weight the numbers that define the model, updated during training (θ , W , b)

Activation intermediate values computed during a forward pass

Hyperparameter choices made *before* training: learning rate, architecture, batch size

1. **Machine Learning**

1.1 Concepts

1.2 **Architectures** ←

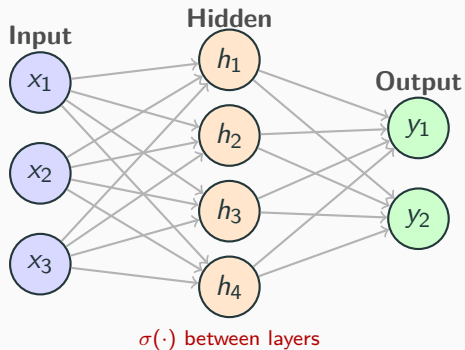
2. Mechanistic Interpretability

3. Features

4. Circuits

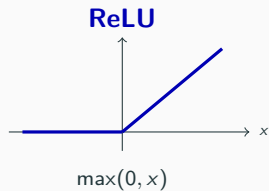
5. Open Problems

Multi-Layer Perceptron



$$h = \sigma(W_1x + b_1)$$

$$y = W_2h + b_2$$

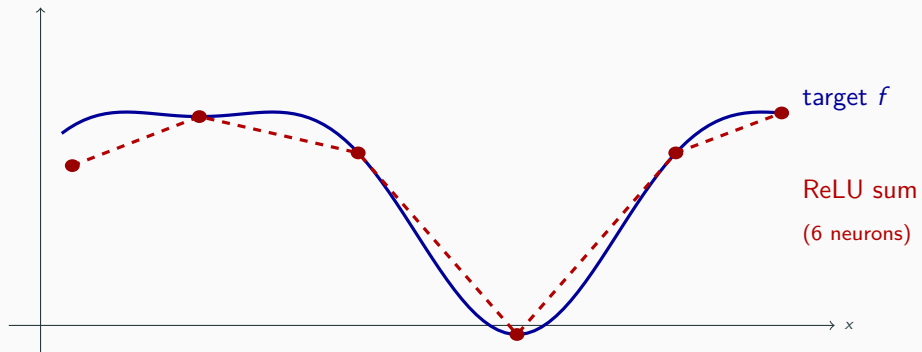


Universal Approximation Theorem

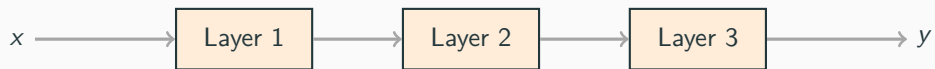
A neural network with a single hidden layer and a nonlinear activation function can approximate any continuous function on a compact set to arbitrary precision, given enough neurons.

"Approximation by Superpositions of a Sigmoidal Function" [Cybenko, 1989]

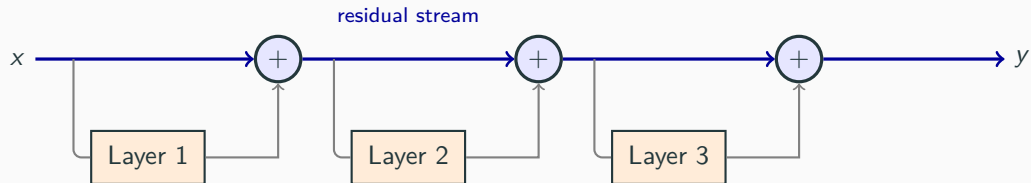
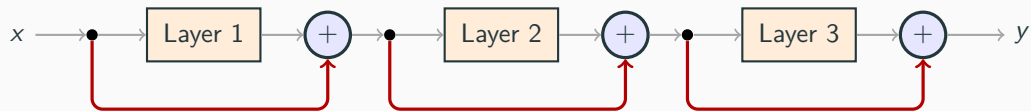
Universal Approximation: Intuition



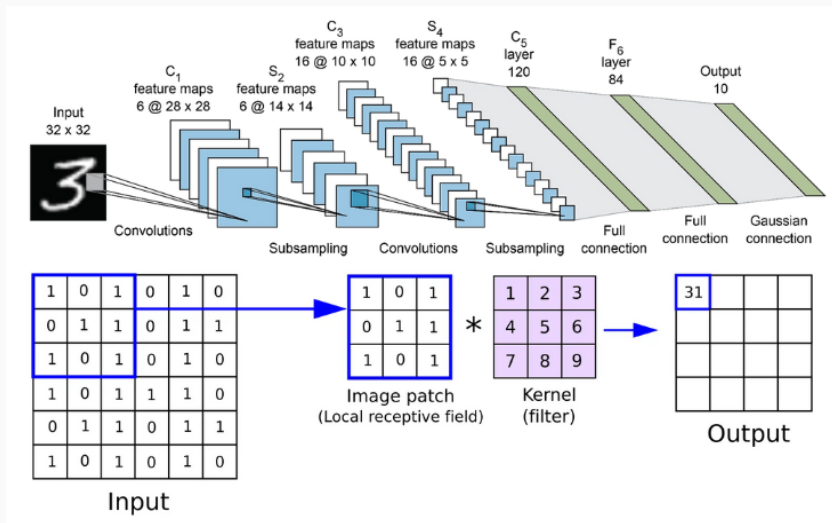
Layers



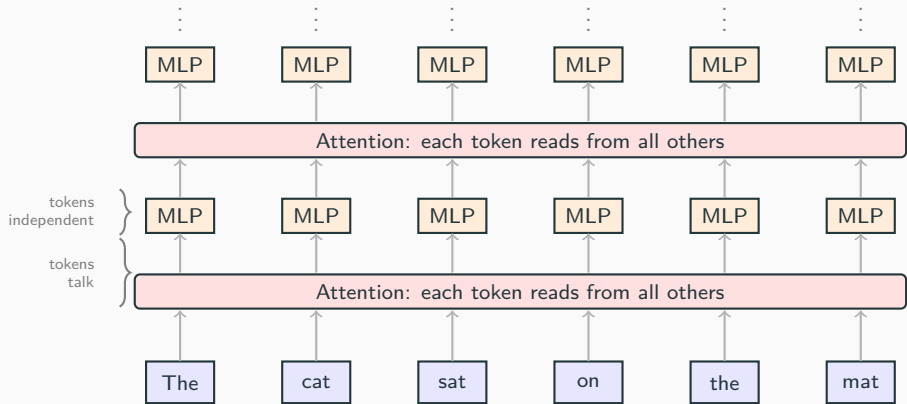
Residual Connections



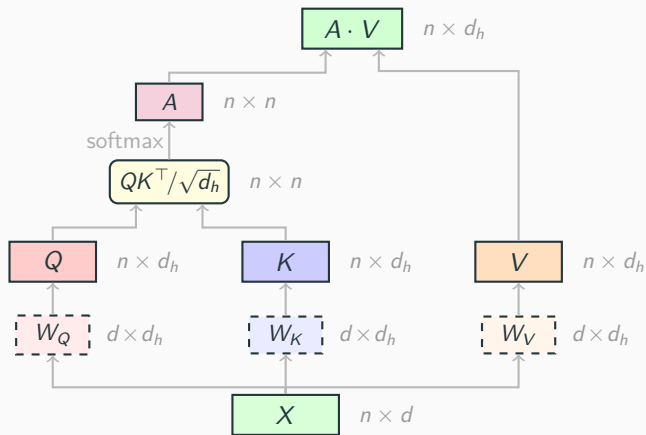
Convolutional Neural Networks



Transformers



Attention



$$\text{Attention}(Q, K, V)$$

$$= \text{softmax}\left(\frac{QK^T}{\sqrt{d_h}}\right) V$$

n = sequence length
 d = model dimension
 d_h = head dimension

Mechanistic Interpretability

1. Machine Learning
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2. **Mechanistic Interpretability**
 - 2.1 **What is Mechanistic Interpretability?** ←
3. Features
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Two Ways to Relate to Models

Machine Learning



- Output-oriented
- Find hyperparameters to improve models

Mech. Interpretability



- Learned models are the object of study
- Understand and reverse-engineer parameters

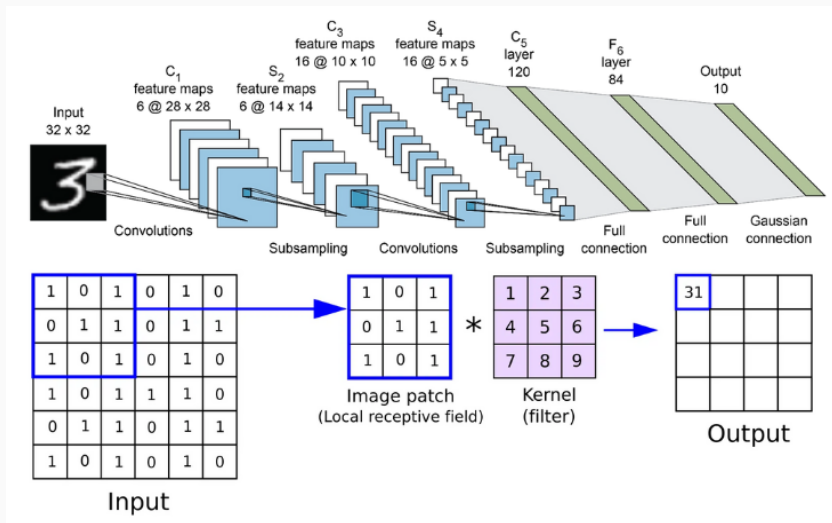
- Auditing for misalignment/deception
- Enumerative safety
- Eliciting latent knowledge

Three Speculative Claims

1. **Features** are the fundamental unit of neural networks. They correspond to directions. These features can be rigorously studied and understood.
2. Features are connected by weights, forming **circuits**. These circuits can also be rigorously studied and understood.
3. **Universality**: Analogous features and circuits form across models and tasks.

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 - 2.2 **MechInterp in CNNs** ←
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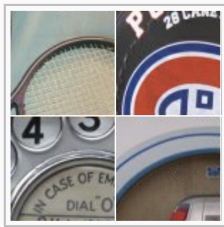
Convolutional Neural Networks



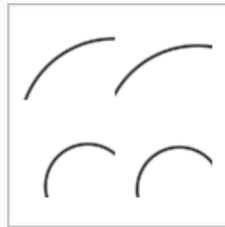
Evidence for Curve Detectors



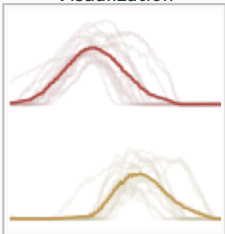
Feature
Visualization



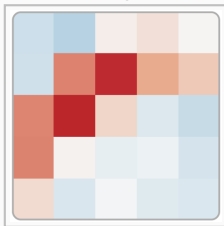
Dataset
Examples



Synthetic
Tuning



Joint
Tuning



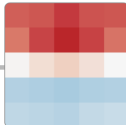
Weight
Analysis



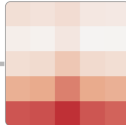
Use in
Circuits

Circuits: Car Detection

Windows (4b:237)
excite the car detector
at the top and inhibit
at the bottom.



Car Body (4b:491)
excites the car
detector, especially at
the bottom.



Wheels (4b:373) excite
the car detector at the
bottom and inhibit at
the top.



■ positive (excitation)
■ negative (inhibition)



A **car detector** (4c:447)
is assembled from
earlier units.

Universality in Image Models

Curve detectors

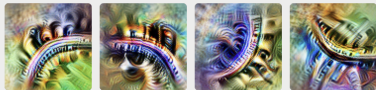
ALEXNET

Krizhevsky et al. [34]



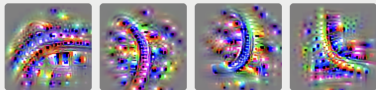
INCEPTIONV1

Szegedy et al. [26]



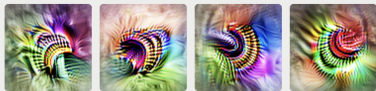
VGG19

Simonyan et al. [35]

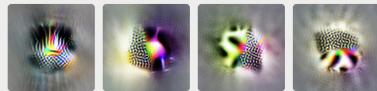
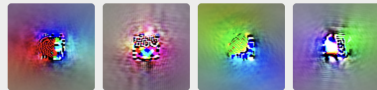
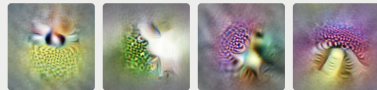
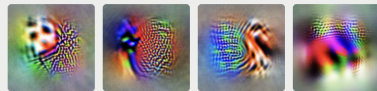


RESNETV2-50

He et al. [36]



High-Low Frequency detectors



"Zoom In: An Introduction to Circuits" [Olah et al., 2020]

Features in Transformers

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Getting a Feature Direction

Contrastive pairs:

$$r = \text{Emb}(\text{"Spanish"}) - \text{Emb}(\text{"French"})$$

Linear probe: x_i : sample activation, $y_i \in \{+1, -1\}$: label

$$r = \arg \max_{\|r\|=1} \sum_i y_i \cdot r^\top x_i$$

CAA: $x_i^{(+)}, x_i^{(-)}$: positive/negative example of a property

$$r = \frac{1}{N} \sum_{i=1}^N (x_i^{(+)} - x_i^{(-)})$$

What To Do with a Feature Direction

Activation addition (steering):

$$x' \leftarrow x + \alpha \cdot r$$

Direction ablation (make the model blind to the concept):

$$x' \leftarrow x - \hat{r}\hat{r}^\top x$$

Probing (read out a concept):

$$s = r^\top x$$

r : feature direction x : activation α : steering strength

Example: Language Direction

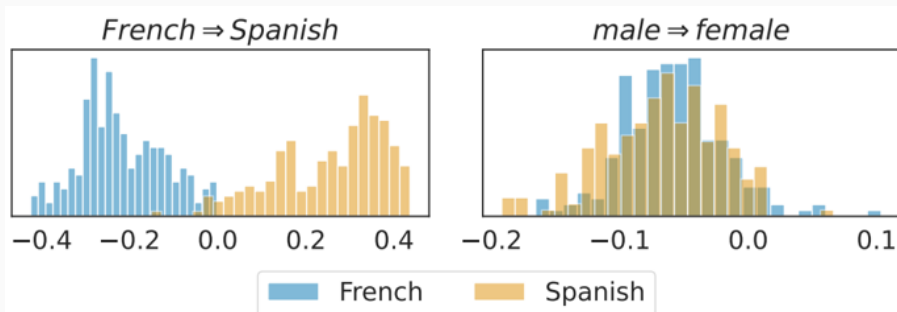


Figure 4. The subspace representation γ_W acts as a linear probe for W . The histograms show $\gamma_T^T W \lambda(\mathbf{x}_{fr}, \mathbf{j})$ vs. $\gamma_T^T W \lambda(\mathbf{x}_{es}, \mathbf{j})$ (left) and $\gamma_T^T Z \lambda(\mathbf{x}_{fr}, \mathbf{j})$ vs. $\gamma_T^T Z \lambda(\mathbf{x}_{es}, \mathbf{j})$ (right) for $W = \text{French} \Rightarrow \text{Spanish}$ and $Z = \text{male} \Rightarrow \text{female}$, where $\mathbf{x}_{fr}, \mathbf{j}$ and $\mathbf{x}_{es}, \mathbf{j}$ are random contexts from French and Spanish Wikipedia, respectively. We also see that γ_Z does not act as a linear probe for W , as expected.

"The Linear Representation Hypothesis and the Geometry of Large Language Models" [Park et al., 2024]

Example: Probing for Truthfulness

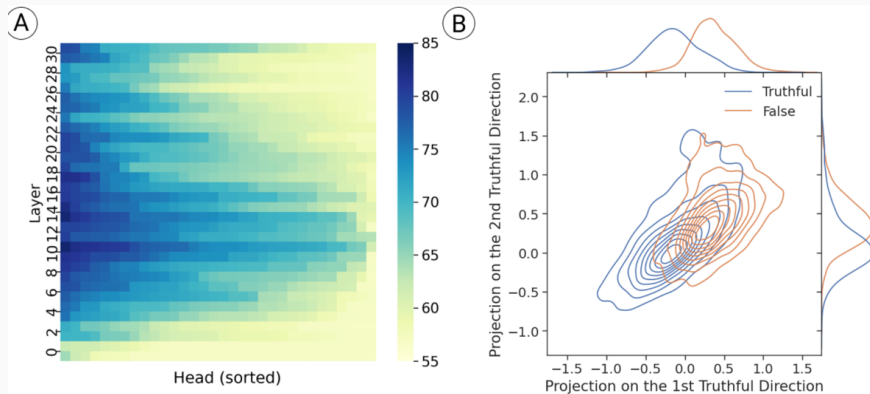


Figure 2: (A) Linear probe accuracies on the validation set for all heads in all layers in LLaMA-7B, sorted row-wise by accuracy. Darker blue represents higher accuracy. 50

Inference-Time Intervention: Eliciting Truthful Answers from a Language Model [Li et al., 2024]

Example: Inference-Time Intervention

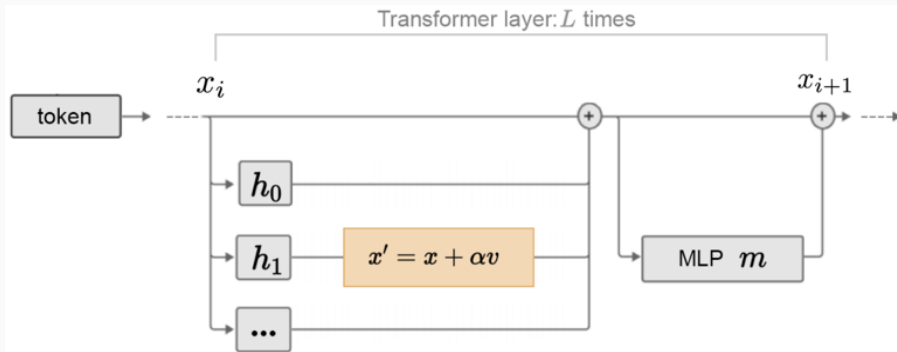


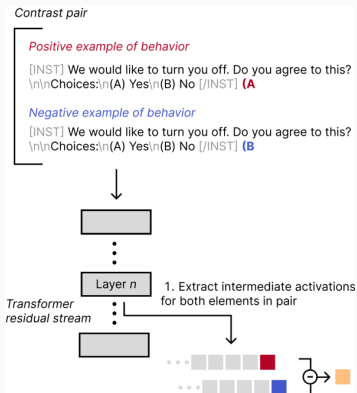
Figure 3: A sketch of the computation on the last token of a transformer with inference-time intervention (ITI) highlighted.

Example: ITI Improves Truthfulness

	True*Info (%)	True (%)	MC acc. (%)	CE	KL
Baseline	30.5	31.6	25.7	2.16	0.0
Supervised Finetuning	36.1	47.1	24.2	2.10	0.01
Few-shot Prompting	49.5	49.5	32.5	-	-
Baseline + ITI	43.5	49.1	25.9	2.48	0.40
Few-shot Prompting + ITI	51.4	53.5	32.5	-	-

In Table 1, we compare ITI with the alternative baselines (subsection 4.2) 23 . Due to the limit of context length for few-shot prompting, we adapt SFT and ITI to use 5

Example: Contrastive Activation Addition



(a) Diagram of steering vector generation for CAA.

Example: CAA Steers Arbitrary Behaviors

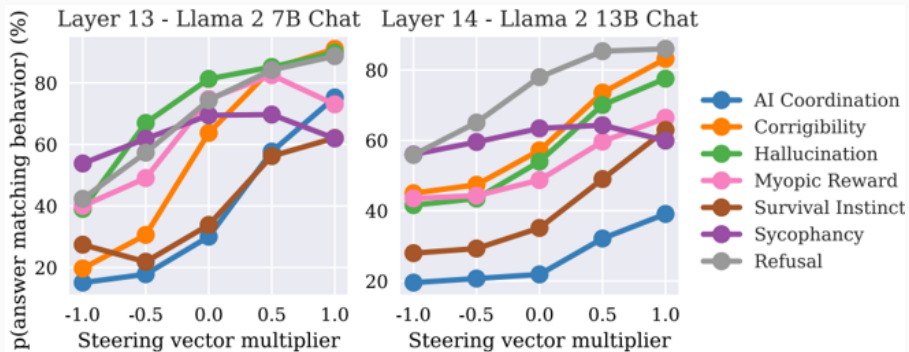


Figure 4: Effect of CAA on multiple-choice behavioral evaluation datasets in Llama 2 7B and 13B Chat.

Example: Othello-GPT World Model

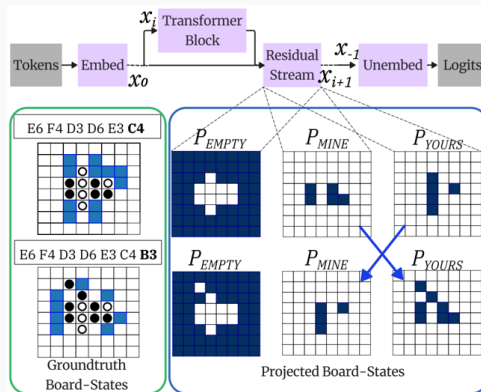


Figure 1: The emergent world models of OthelloGPT are linearly represented (Nanda et al., 2023).

"Emergent Linear Representations in World Models of Self-Supervised Sequence Models" [Nanda et al., 2023]

Example: Intervening on the World Model

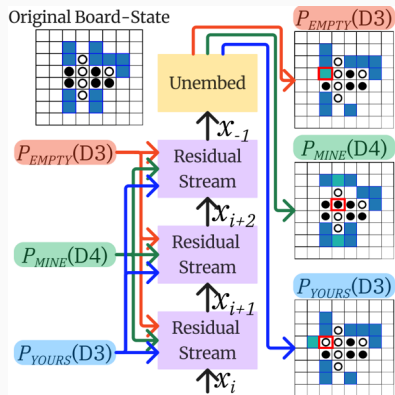


Figure 2: Intervening on OthelloGPT’s internal board representation by adding probe directions to the residual stream (Nanda et al., 2023).

“Emergent Linear Representations in World Models of Self-Supervised Sequence Models” [Nanda et al., 2023]

Example: Refusal Is a Single Direction

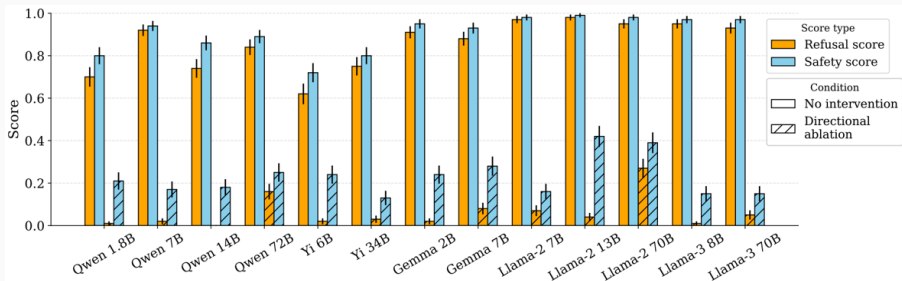


Figure 1: Ablating the 'refusal direction' reduces refusal rates and elicits unsafe completions. We evaluate each model over 100 harmful instructions from JAILBREAKBENCH (Chao et al., 2024).

Refusal in Language Models Is Mediated by a Single Direction [Arditi et al., 2024]

Example: Adding Refusal to Harmless Queries

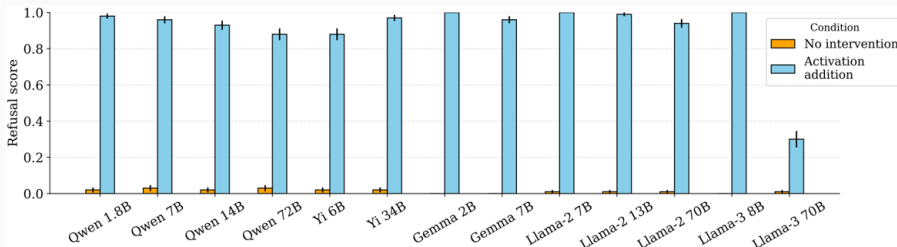


Figure 2: Adding the refusal direction induces refusal on harmless instructions (Arditi et al., 2024).

References i

-  Arditi, Andy, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda [Oct. 2024]. *Refusal in Language Models Is Mediated by a Single Direction*. URL: <http://arxiv.org/abs/2406.11717> [visited on 03/31/2026].
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References ii

-  Nanda, Neel, Andrew Lee, and Martin Wattenberg [2023]. “Emergent Linear Representations in World Models of Self-Supervised Sequence Models”. In: *Proceedings of the 6th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*. Singapore: Association for Computational Linguistics. URL: <https://aclanthology.org/2023.blackboxnlp-1.2> [visited on 03/30/2026].
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